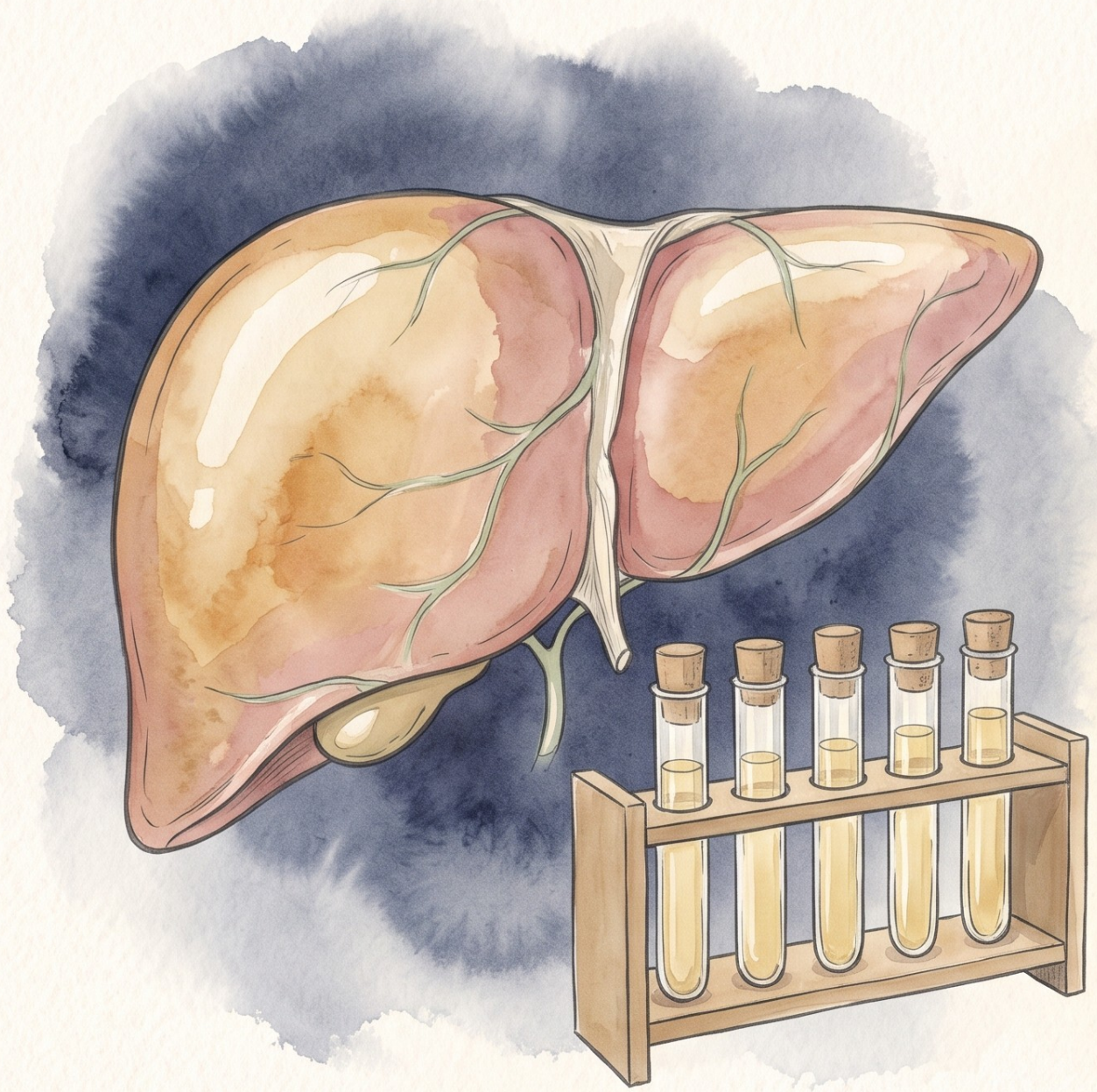


Chronic Hepatitis B

Mohammad Almeqdadi, MD
Transplant Hepatologist and Gastroenterologist
Assistant Professor of Medicine and Surgery
X: @mfalmeqdadi

Tufts UNIVERSITY | School of Medicine

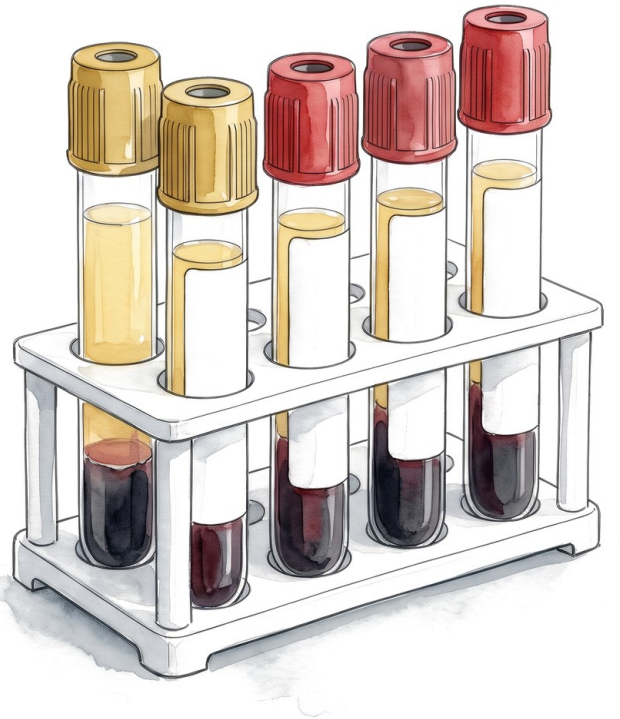
Slides (PDF)

Illustrations generated with Google Gemini

Objectives

By the end of this talk you should be able to:

- Read a hepatitis B serology panel
- Place a patient in the right phase
- Decide who to treat, using the current guidelines
- Estimate HCC risk and set up surveillance
- Prevent reactivation and transmission to the baby



The Australia antigen

A "New" Antigen in Leukemia Sera

The "Australia antigen" is found in the sera of some normal individuals from foreign populations. The total absence of the antigen from the sera of normal United States subjects and its relatively high frequency in acute leukemia suggests that the presence of the antigen may be of value in the diagnosis of early acute leukemia. Whether the antigen results from or precedes the leukemia process remains to be seen.

In 1963, sera from patients with hemophilia who had received transfusions were tested for the presence of isoprecipitins using a panel of 24 sera from normal individuals, including sera from foreign populations. Two of the hemophilia sera formed a clearly defined precipitin line with one of the panel sera (from an Australian aborigine), but with none

- Found in the serum of some normal people from foreign populations, and in none of the normal US subjects
- Frequent in acute leukemia, which is why the paper is about leukemia
- Blumberg shared the 1976 Nobel Prize with Carleton Gajdusek

Blumberg BS, Alter HJ, Visnich S. JAMA 1965;191:541-546. PMID 14239025.

From antigen to vaccine

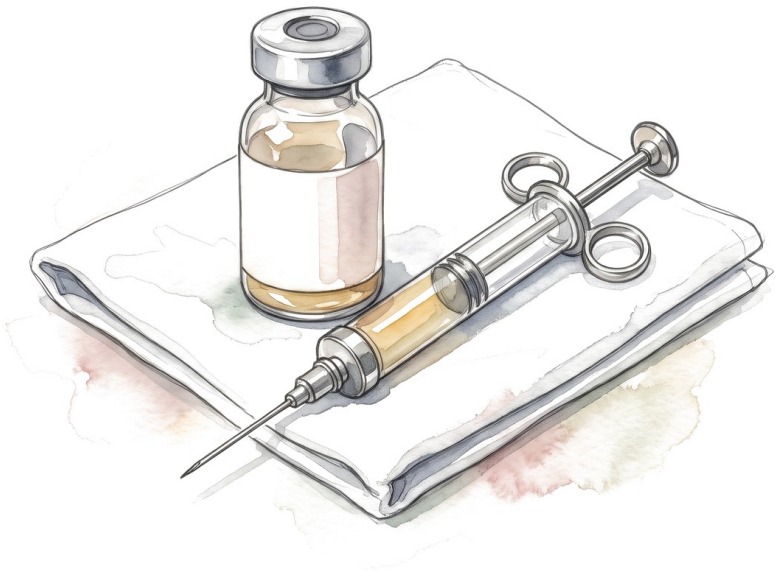
1965 — Blumberg, Alter and Visnich describe the Australia antigen

1970 — Dane, Cameron and Briggs see the virus particle in serum

1976 — Blumberg and Gajdusek receive the Nobel Prize. First immunization of people with plasma-derived vaccine (Maupas)

1980 — Szmuness: the vaccine works in a controlled trial of 1,083 men

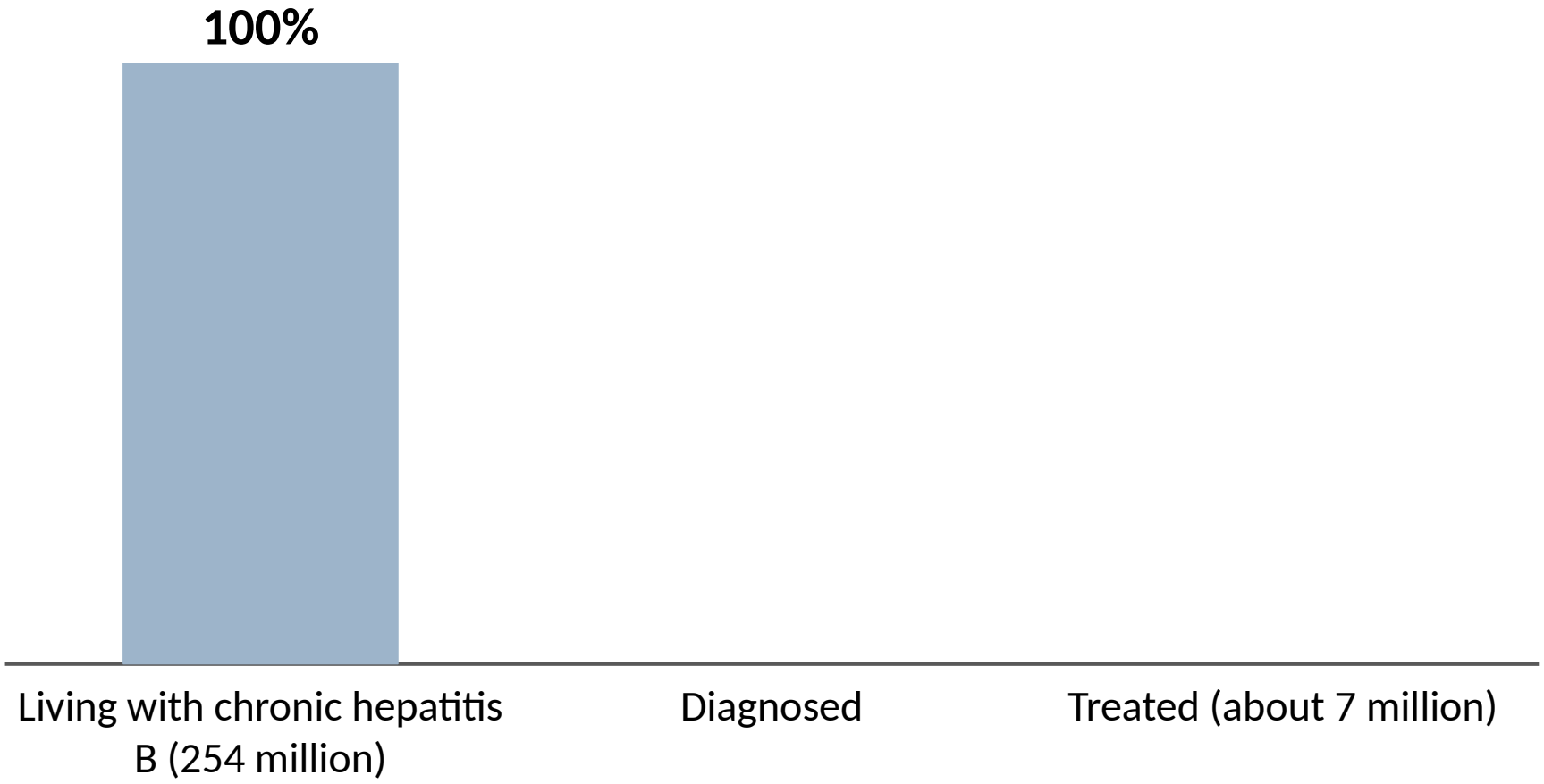
1981 — Beasley: HBsAg carriers in Taiwan get liver cancer at a far higher rate



PMIDs: Dane 4190997; Maupas 59012; Szmuness 6997738; Beasley 6118576.

Hepatitis B worldwide

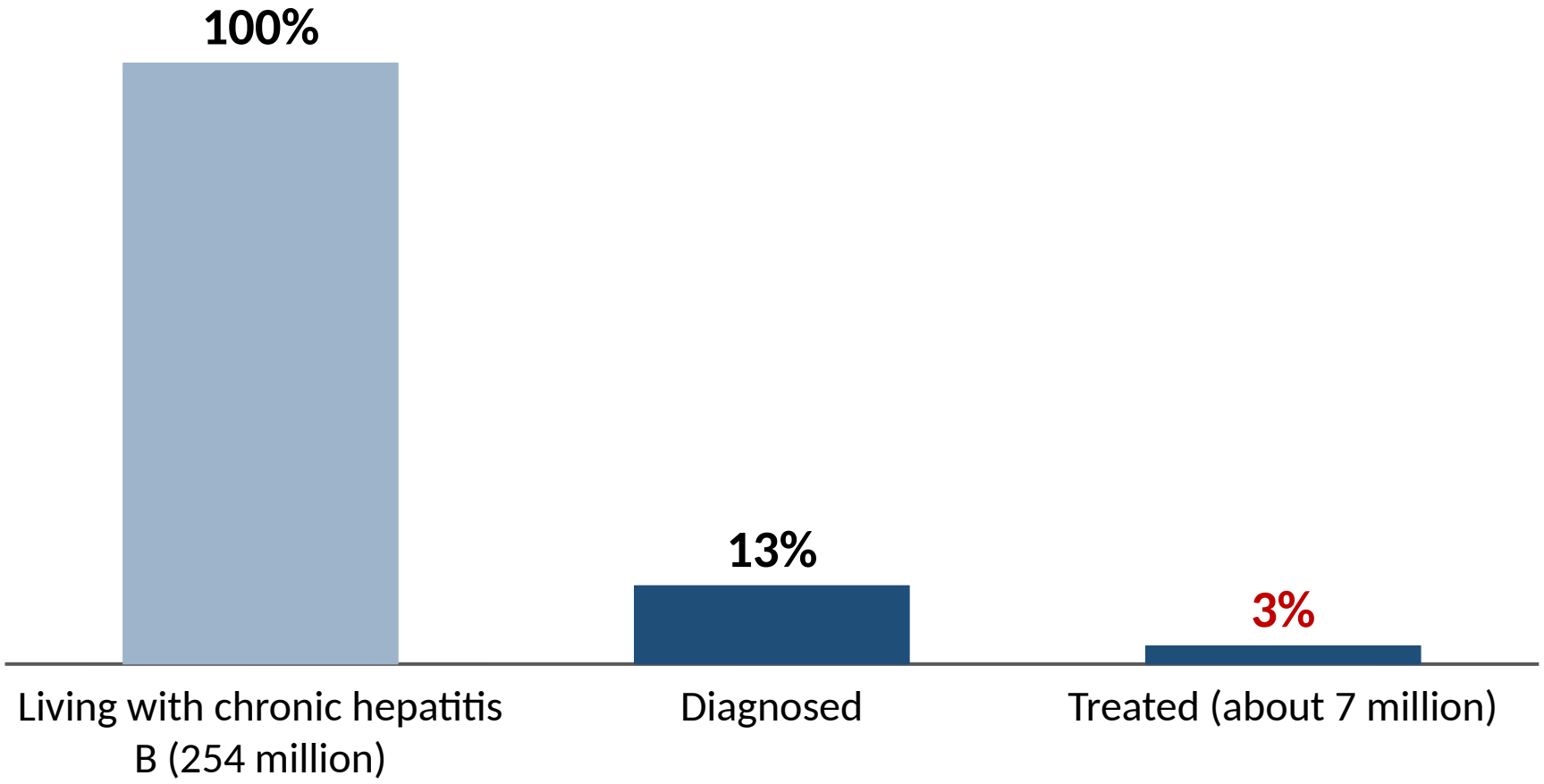
How many people with chronic hepatitis B know it?



WHO news release 9 April 2024 (2022 estimates), rounded. WHO Global hepatitis report 2024.

Hepatitis B worldwide

How many people with chronic hepatitis B know it?



WHO news release 9 April 2024 (2022 estimates), rounded. WHO Global hepatitis report 2024.

Screening an adult for hepatitis B

Which three tests make up the screening panel?

Question	Test
Is there a current infection?	
Is the patient immune?	
Has the patient ever been exposed?	

CDC recommendations, MMWR Recomm Rep 2023;72(1):1-25, PMID 36893044.

Screening an adult for hepatitis B

Which three tests make up the screening panel?

Question	Test
Is there a current infection?	HBsAg
Is the patient immune?	Anti-HBs
Has the patient ever been exposed?	Total anti-HBc

Once for every adult 18 and older, and in every pregnancy

CDC recommendations, MMWR Recomm Rep 2023;72(1):1-25, PMID 36893044.

Five panels

What does each panel mean?

	HBsAg	Total anti-HBc	IgM anti-HBc	Anti-HBs	Interpretation
Panel 1	–	–	–	–	
Panel 2	–	–	–	+	
Panel 3	–	+	–	+	
Panel 4	+	+	–	–	
Panel 5	+	+	+	–	

Interpretation per CDC 2023 (Table 1), MMWR Recomm Rep 2023;72(1):1-25.

Five panels

What does each panel mean?

	HBsAg	Total anti-HBc	IgM anti-HBc	Anti-HBs	Interpretation
Panel 1	–	–	–	–	Susceptible
Panel 2	–	–	–	+	Vaccinated
Panel 3	–	+	–	+	Resolved infection
Panel 4	+	+	–	–	Chronic infection
Panel 5	+	+	+	–	Acute infection

Interpretation per CDC 2023 (Table 1), MMWR Recomm Rep 2023;72(1):1-25.

61-year-old woman starting rituximab

HBsAg	Total anti-HBc	Anti-HBs	HBV DNA	ALT
–	+	–	Not done	28 U/L

What could this be, and what do you order?

Possibility	How you sort it out

CDC 2023 (Table 1), PMID 36893044; AASLD 2018 hepatitis B guidance, PMID 29405329.

61-year-old woman starting rituximab

HBsAg	Total anti-HBc	Anti-HBs	HBV DNA	ALT
–	+	–	Not done	28 U/L

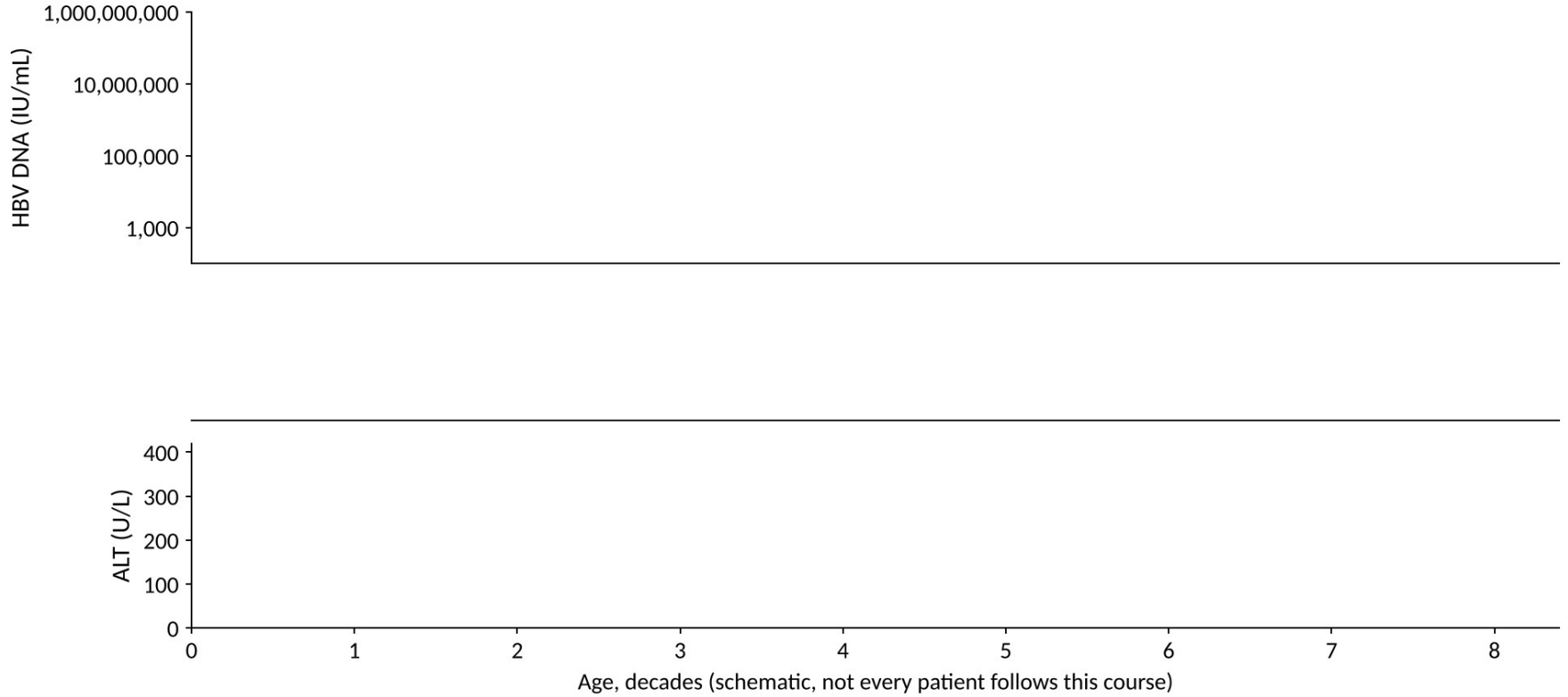
What could this be, and what do you order?

Possibility	How you sort it out
Resolved infection, anti-HBs has waned	HBV DNA undetectable
Occult infection	HBV DNA detectable
Window period	IgM anti-HBc positive
False-positive anti-HBc	Repeat testing, DNA undetectable

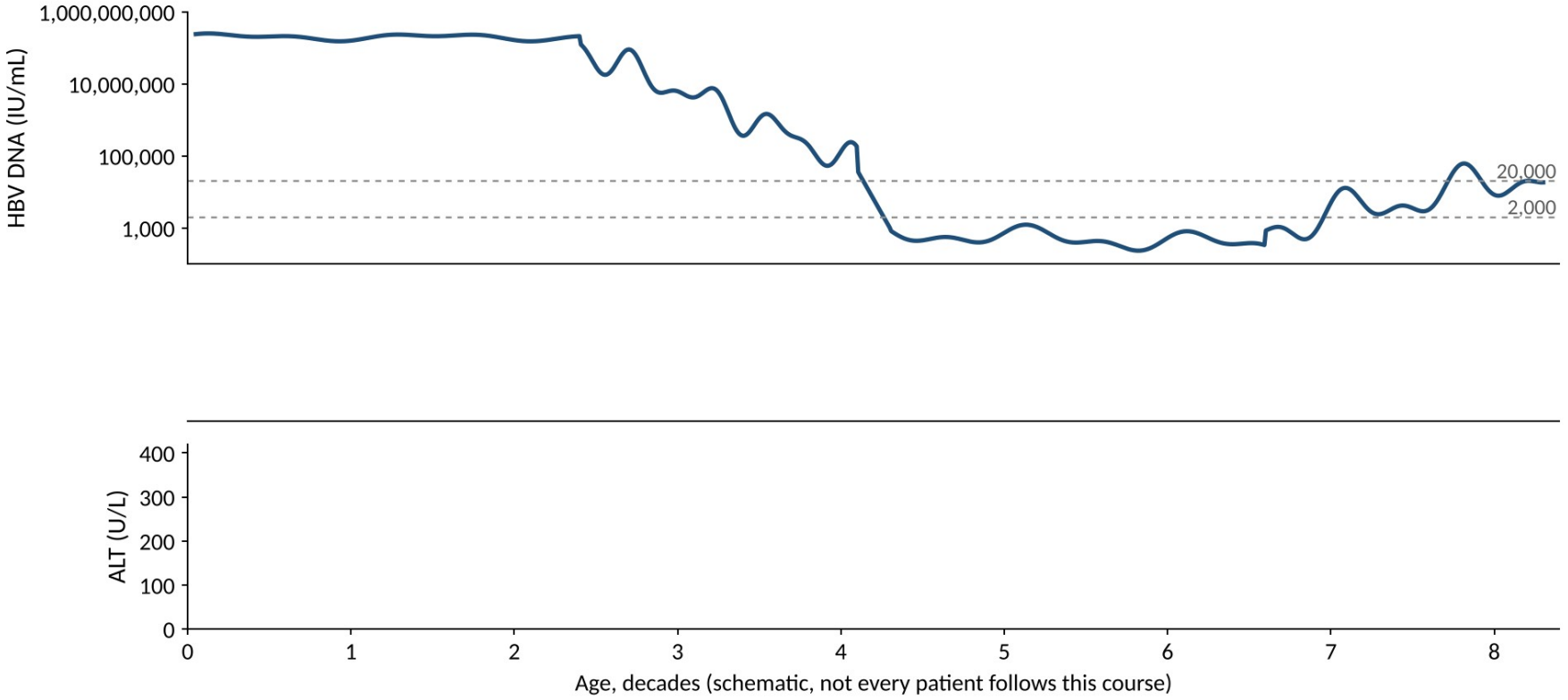
CDC 2023 (Table 1), PMID 36893044; AASLD 2018 hepatitis B guidance, PMID 29405329.

Natural history of chronic hepatitis B

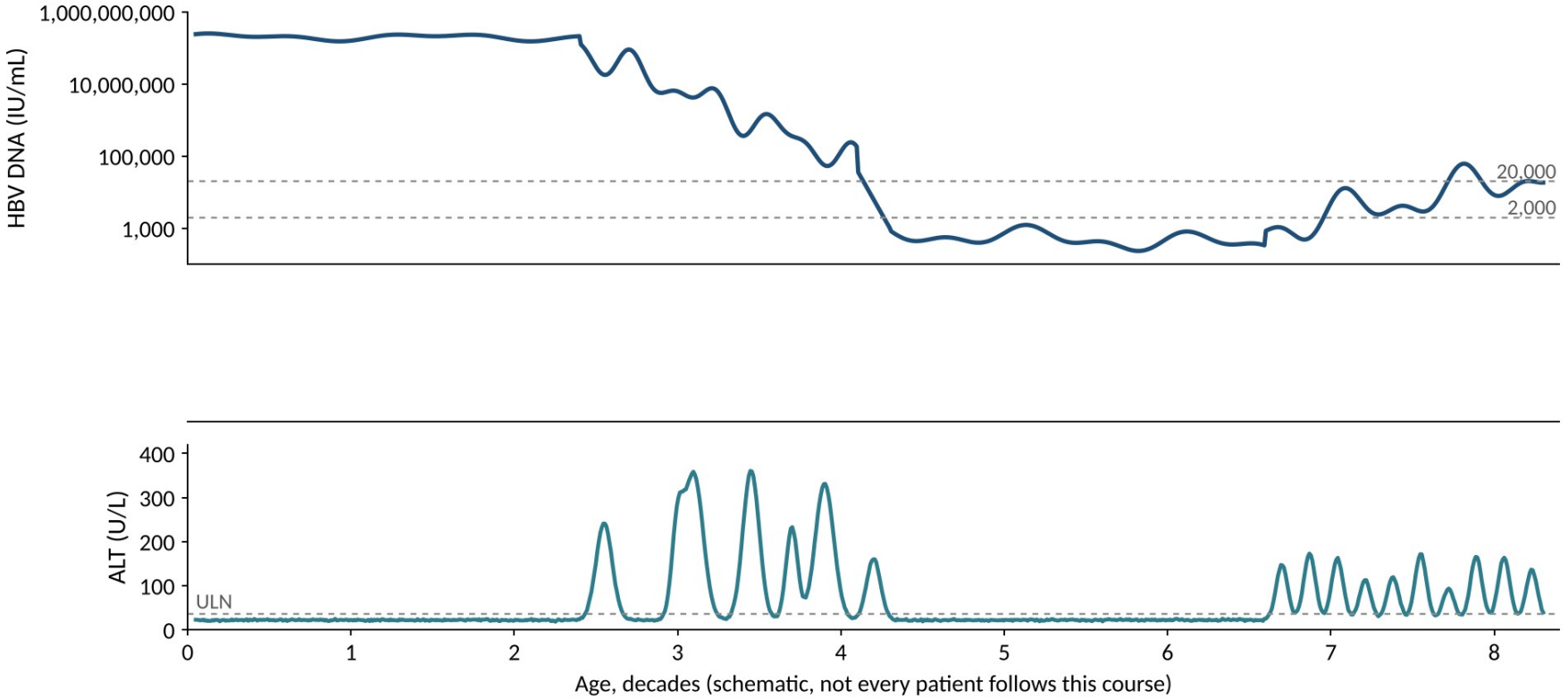
Where do HBV DNA, ALT and the antigens go over a lifetime?



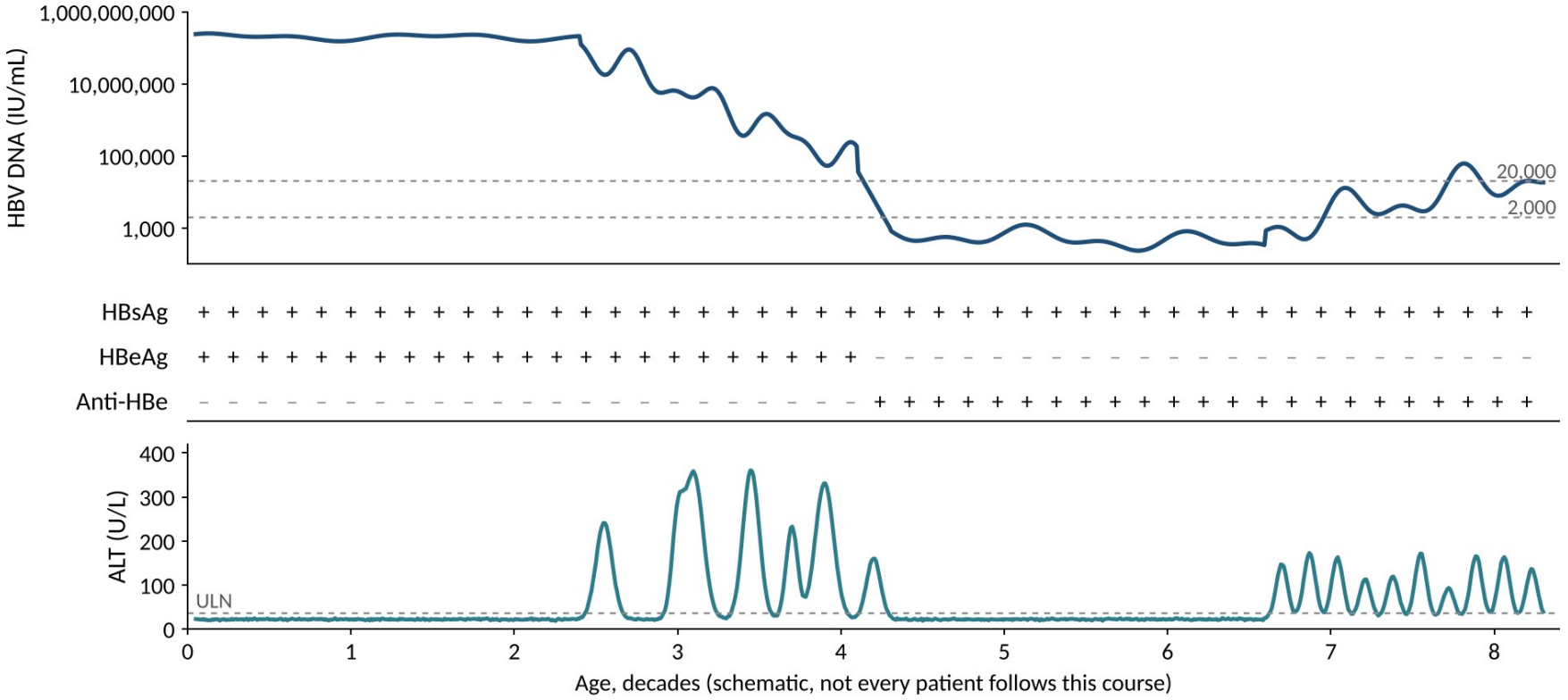
Natural history of chronic hepatitis B



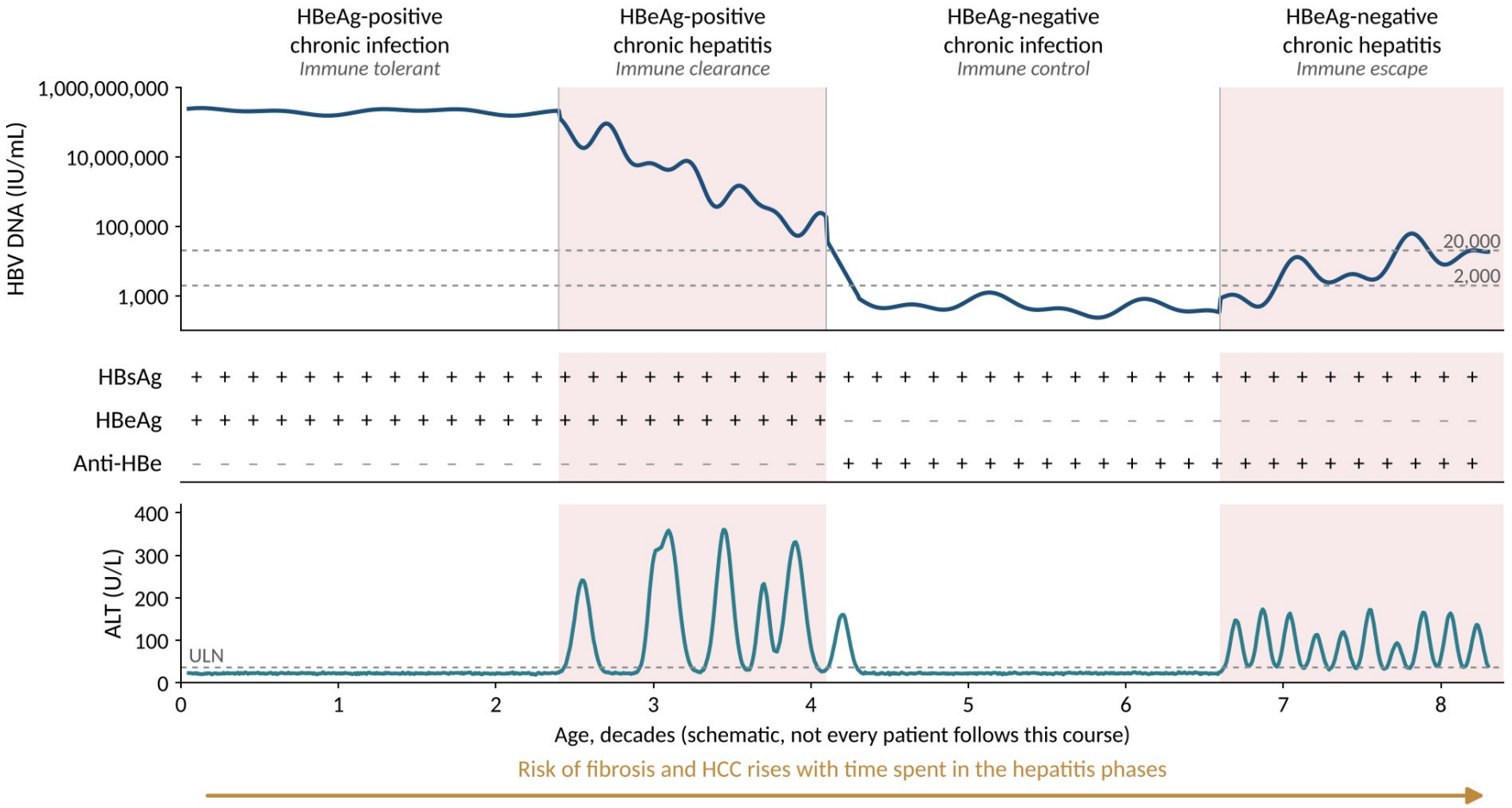
Natural history of chronic hepatitis B



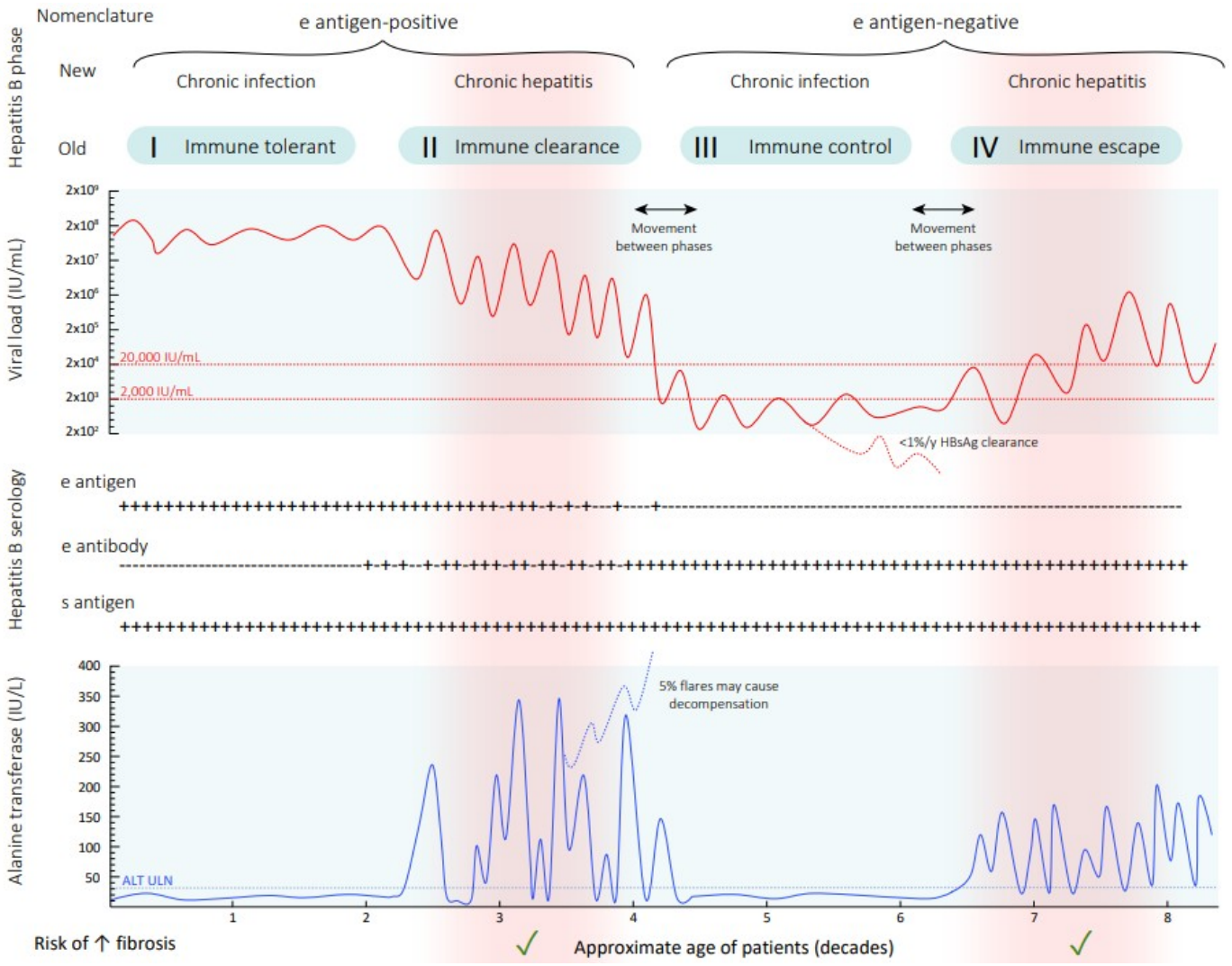
Natural history of chronic hepatitis B



Natural history of chronic hepatitis B



The published version



GESA Australian consensus recommendations for the management of hepatitis B infection, 2022, Figure 3.1 (as shown in the ASHM B Positive chapter).

Match the labs to the phase

HBeAg	HBV DNA (IU/mL)	ALT	Phase
Positive	10,000,000 or more	Normal	
Positive	10,000 to 10,000,000	Raised	
Negative	Below 2,000	Normal	
Negative	Above 2,000	Raised	

EASL Clinical Practice Guidelines on hepatitis B 2025, Table 4. J Hepatol 2025;83:502-583, PMID 40348683. Values are 'usually'.

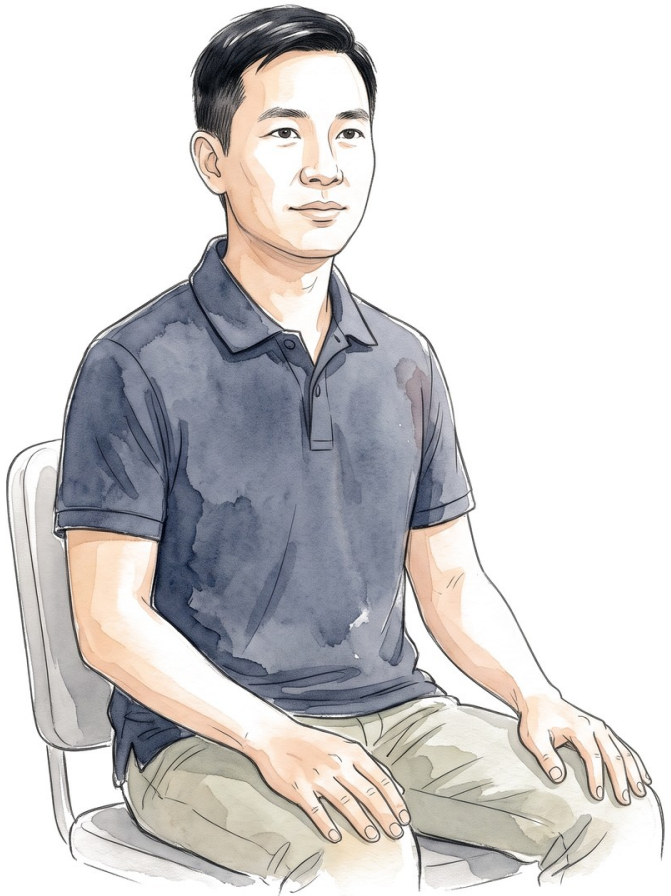
Match the labs to the phase

HBeAg	HBV DNA (IU/mL)	ALT	Phase
Positive	10,000,000 or more	Normal	HBeAg-positive chronic infection
Positive	10,000 to 10,000,000	Raised	HBeAg-positive chronic hepatitis
Negative	Below 2,000	Normal	HBeAg-negative chronic infection
Negative	Above 2,000	Raised	HBeAg-negative chronic hepatitis

EASL Clinical Practice Guidelines on hepatitis B 2025, Table 4. J Hepatol 2025;83:502-583, PMID 40348683. Values are 'usually'.

34-year-old man, born in Vietnam

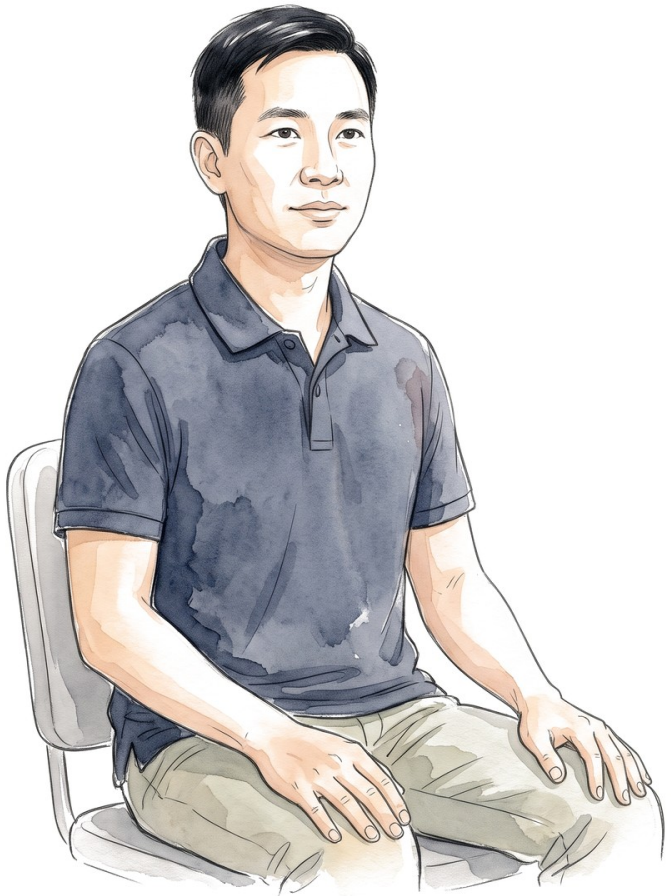
Test	Result
HBsAg	Positive
HBeAg	Positive
Anti-HBe	Negative
HBV DNA	60,000,000 IU/mL
ALT (ULN 35)	24 U/L
Platelets	245 x 10^9/L
Liver stiffness	5.1 kPa



Do you treat him?

34-year-old man, born in Vietnam

Test	Result
HBsAg	Positive
HBeAg	Positive
Anti-HBe	Negative
HBV DNA	60,000,000 IU/mL
ALT (ULN 35)	24 U/L
Platelets	245 x 10^9/L
Liver stiffness	5.1 kPa

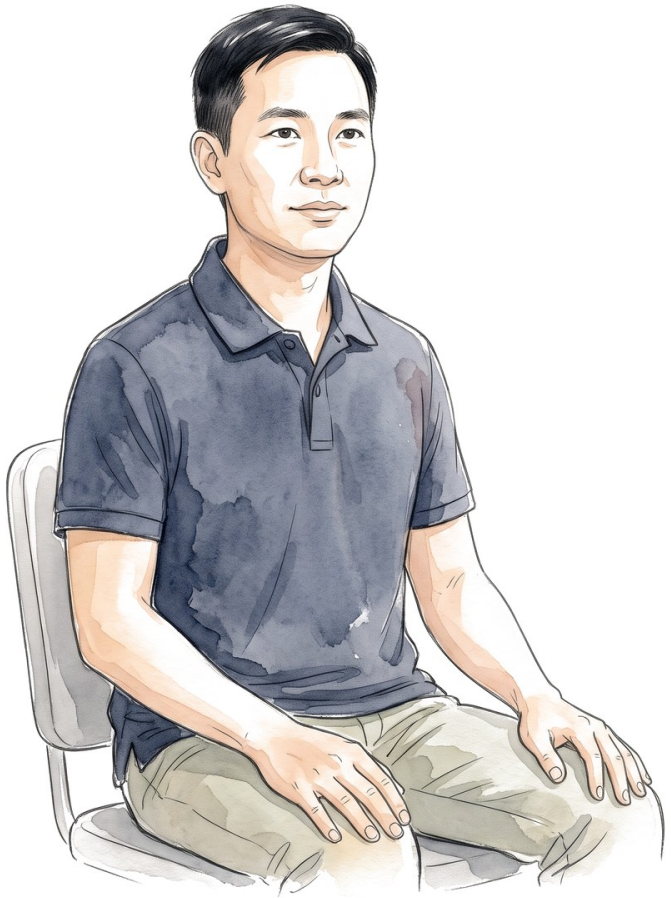


Do you treat him?

No. Monitor ALT and DNA every 3 to 6 months

Same man, now 41 years old

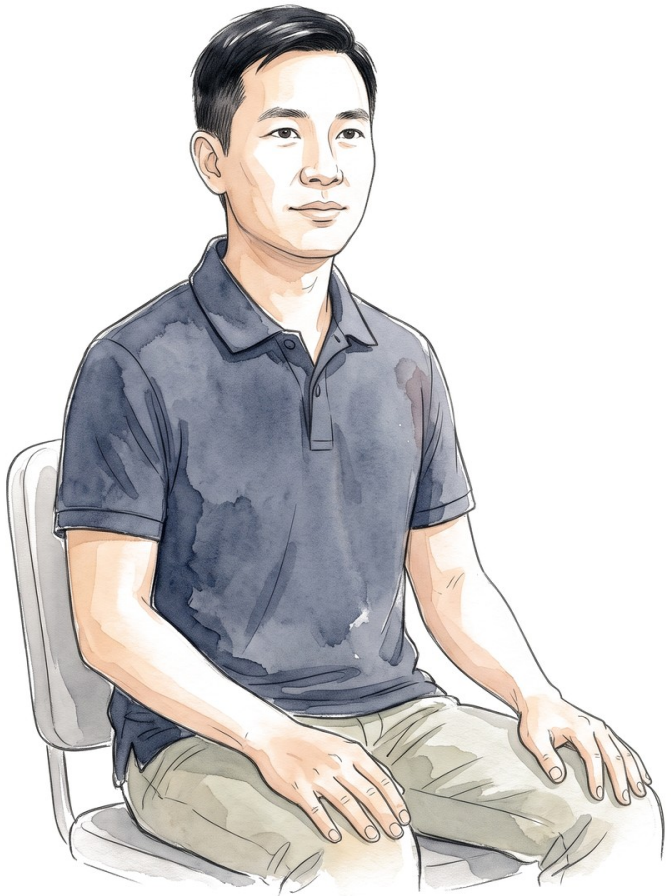
Test	Result
HBsAg	Positive
HBeAg	Positive
Anti-HBe	Negative
HBV DNA	55,000,000 IU/mL
ALT (ULN 35)	26 U/L
Platelets	245 x 10^9/L
Liver stiffness	5.1 kPa



Does the answer change?

Same man, now 41 years old

Test	Result
HBsAg	Positive
HBeAg	Positive
Anti-HBe	Negative
HBV DNA	55,000,000 IU/mL
ALT (ULN 35)	26 U/L
Platelets	245 x 10^9/L
Liver stiffness	5.1 kPa

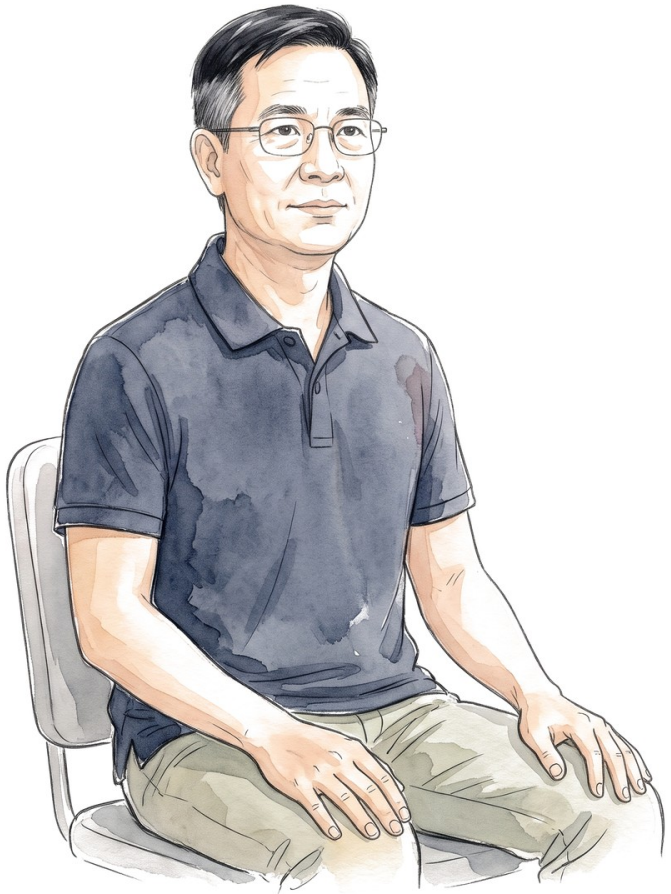


Does the answer change?

AASLD/IDSA 2025: over age 40 supports treatment

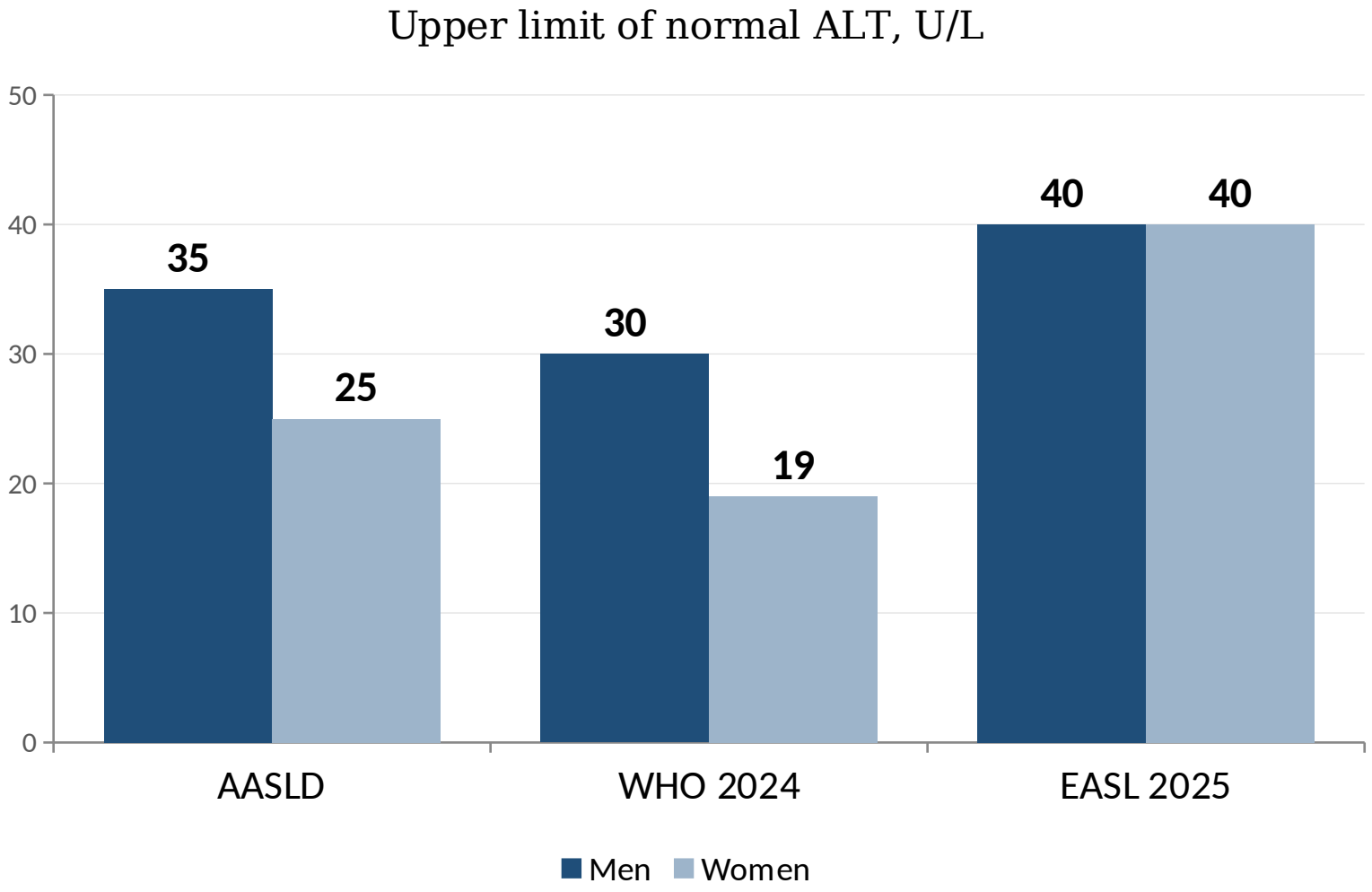
Same man, now 52 years old

Test	Result
HBsAg	Positive
HBeAg	Negative
Anti-HBe	Positive
HBV DNA	8,000 IU/mL
ALT	48 U/L
Platelets	190 x 10^9/L
FIB-4	1.8
Liver stiffness	6.8 kPa



Do you treat him?

What counts as a normal ALT?



His ALT

48 U/L

Above all three limits

Upper limits of normal as listed in the guidelines. AASLD 2018 PMID 29405329; WHO 2024; EASL 2025 PMID 40348683.

Same labs, four guidelines

Treat or not?

	AASLD 2018	AASLD/IDSA 2025	EASL 2025	WHO 2024
52 y, HBeAg-negative, DNA 8,000, ALT 48				
Compensated cirrhosis, DNA 500				
Decompensated cirrhosis, any DNA				

AASLD 2018 PMID 29405329; AASLD/IDSA 2025 Hepatology 2026;83:974-997 PMID 41186418; EASL 2025 PMID 40348683; WHO 2024 (NBK614987).

Same labs, four guidelines

Treat or not?

	AASLD 2018	AASLD/IDSA 2025	EASL 2025	WHO 2024
52 y, HBeAg-negative, DNA 8,000, ALT 48	Observe	Shared decision	**Treat**	**Treat**
Compensated cirrhosis, DNA 500	Treat	Treat	Treat	Treat
Decompensated cirrhosis, any DNA	Treat	Treat	Treat	Treat

AASLD 2018 PMID 29405329; AASLD/IDSA 2025 Hepatology 2026;83:974-997 PMID 41186418; EASL 2025 PMID 40348683; WHO 2024 (NBK614987).

Does treating this group help?

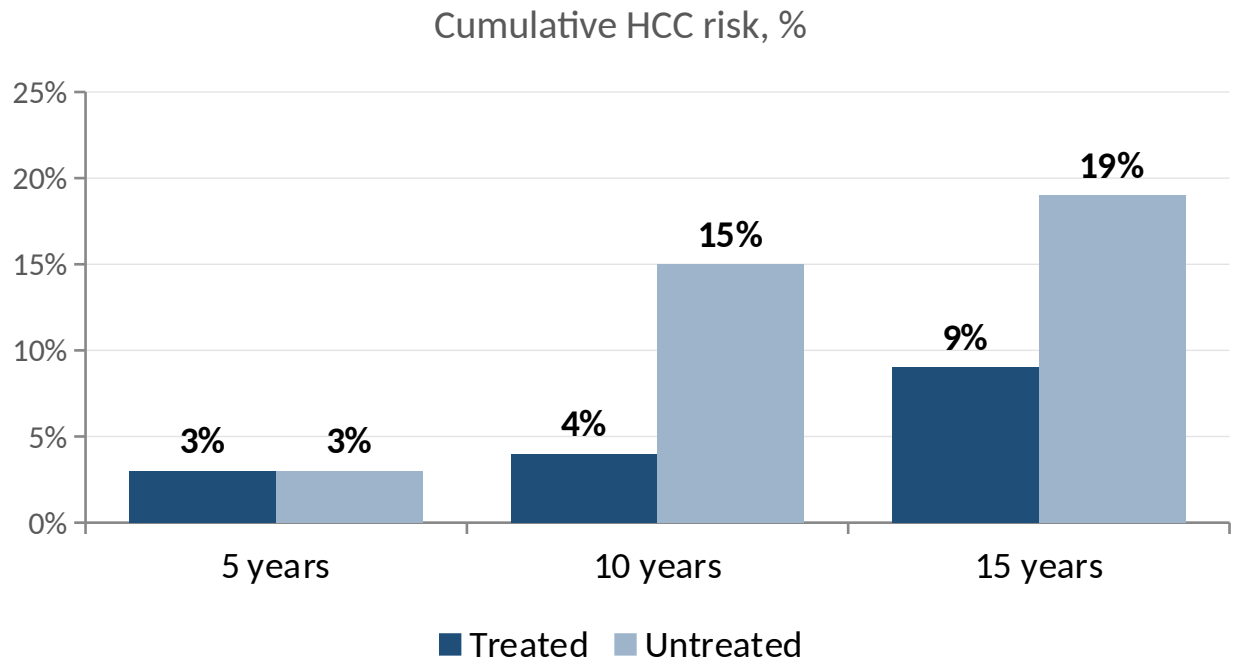
Study	Design	Patients	Outcome
Huang 2023	Observational	855, no advanced fibrosis	HCC, 5 to 15 years
TORCH-B 2021	Randomized	160, ALT 1 to 2 x ULN	Progression at 3 years

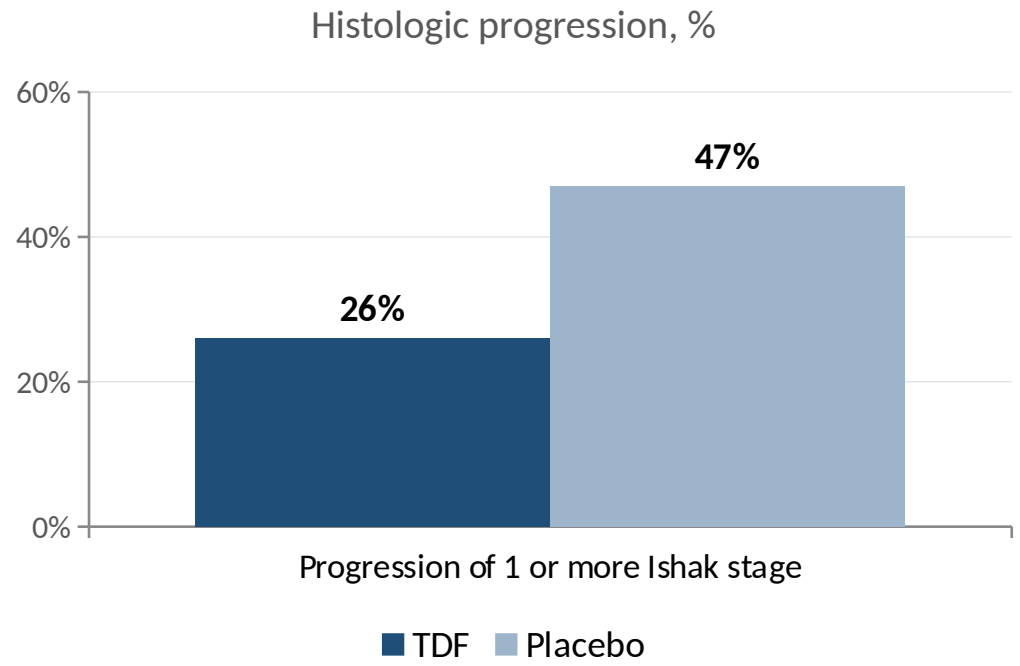
What do you expect?

Huang DQ et al. Hepatology 2023;78:1558-1568, PMID 37184202. Hsu YC et al. Lancet Infect Dis 2021;21:823-833, PMID 33524314.

Does treating this group help?

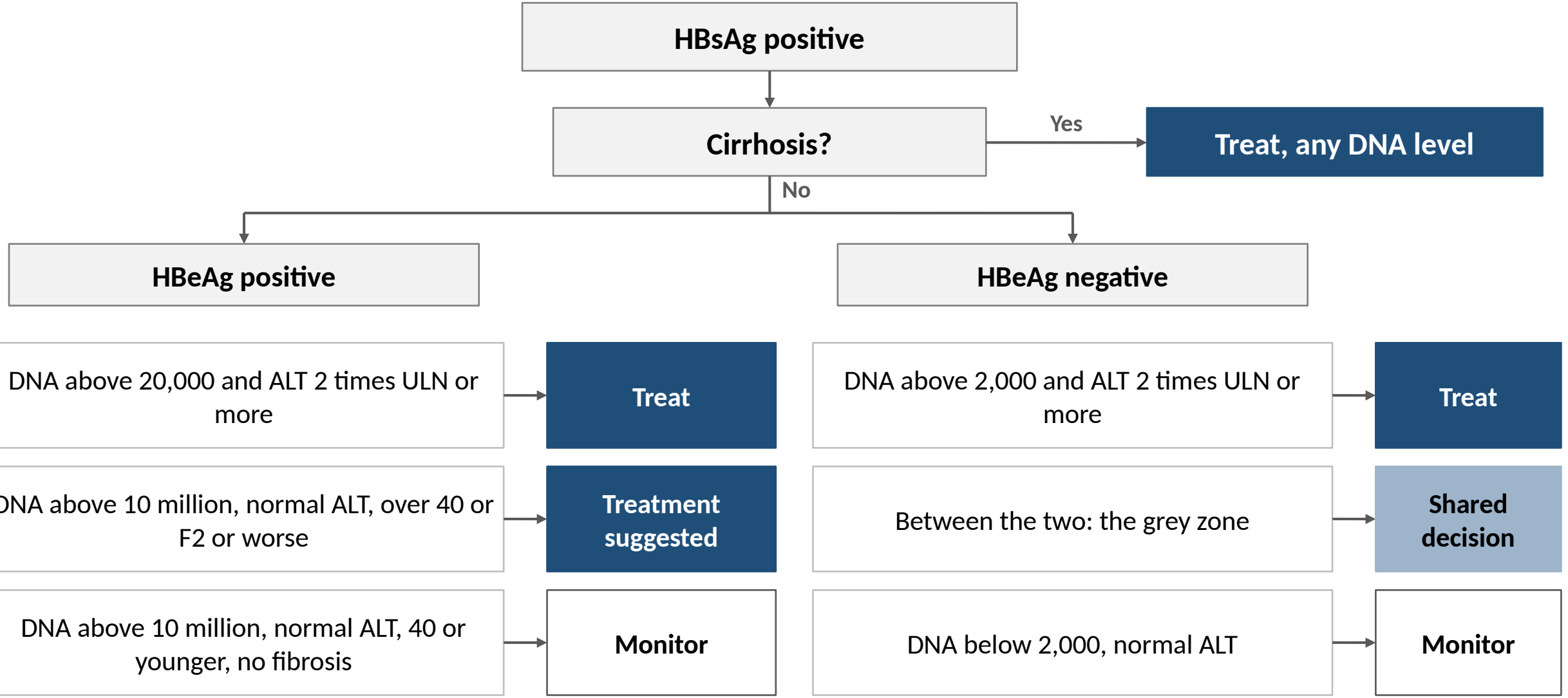
Study	Design	Patients	Outcome
Huang 2023	Observational	855, no advanced fibrosis	HCC, 5 to 15 years
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Huang DQ et al. Hepatology 2023;78:1558-1568, PMID 37184202. Hsu YC et al. Lancet Infect Dis 2021;21:823-833, PMID 33524314.

Who to treat: summary



AASLD 2018 (PMID 29405329); AASLD/IDSA 2025 R3, R4 and Figure 5 (PMID 41186418). ALT ULN 35 men, 25 women.

58-year-old man, eGFR 45

Which nucleos(t)ide analogue?

	Dose	At eGFR 30 to 49	
Entecavir	0.5 mg daily	0.5 mg every 48 hours	
TDF	300 mg daily	300 mg every 48 hours	
TAF	25 mg daily with food	25 mg daily	

US prescribing information: Baraclude, Viread, Vemlidy (DailyMed). Renal doses use creatinine clearance.

58-year-old man, eGFR 45

Which nucleos(t)ide analogue?

	Dose	At eGFR 30 to 49	
Entecavir	0.5 mg daily	0.5 mg every 48 hours	Dose-adjust
TDF	300 mg daily	300 mg every 48 hours	Dose-adjust, kidney and bone effects
TAF	25 mg daily with food	25 mg daily	**No adjustment**

TAF, or entecavir with the dose adjusted

US prescribing information: Baraclude, Viread, Vemlidy (DailyMed). Renal doses use creatinine clearance.

TAF versus TDF at week 48

HBV DNA below 29 IU/mL	TAF	TDF
HBeAg-positive (Chan)	64%	67%
HBeAg-negative (Buti)	94%	93%

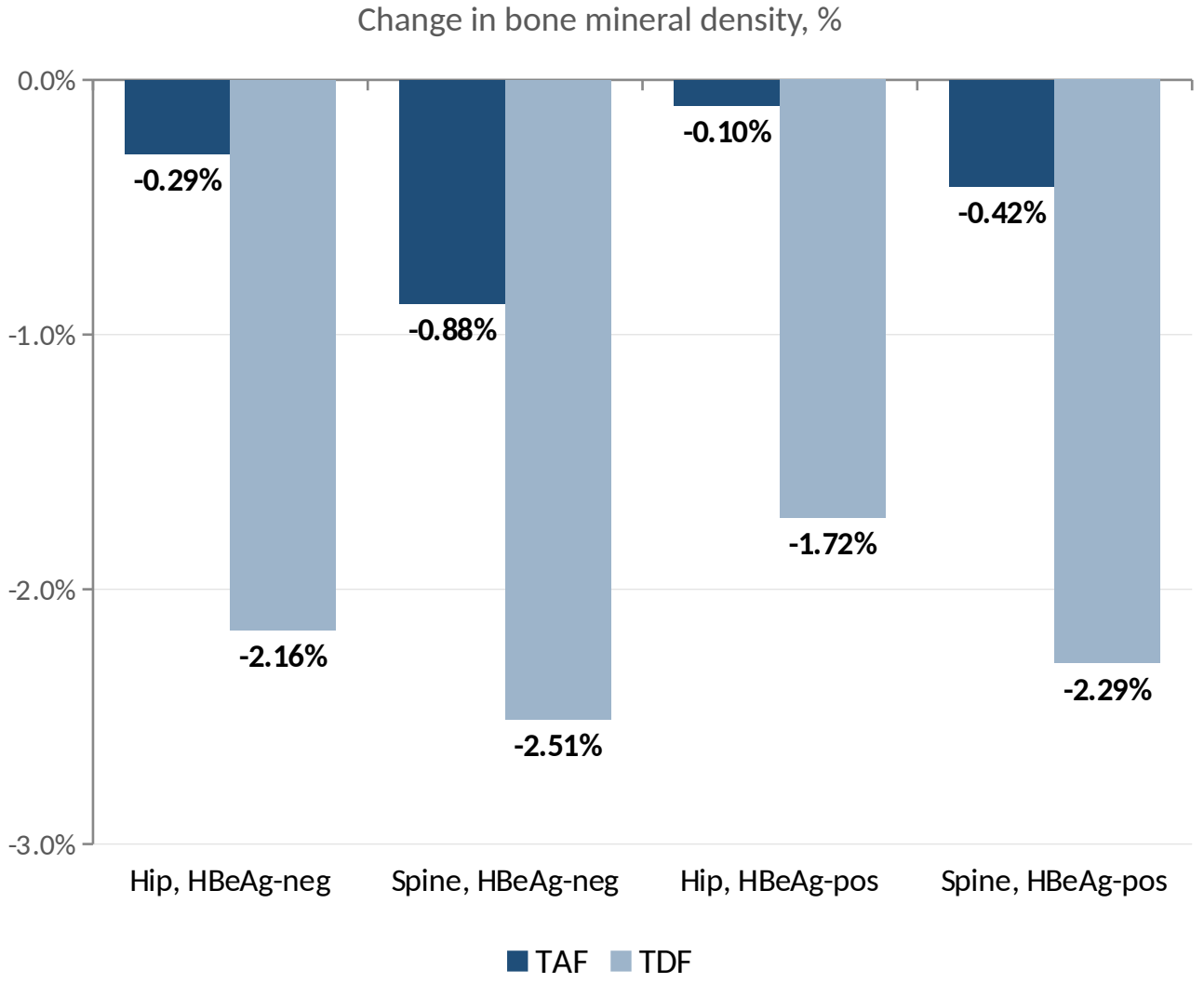
What differs?

Chan HLY et al. Lancet Gastroenterol Hepatol 2016;1:185-195 (PMID 28404091). Buti M et al. 2016;1:196-206 (PMID 28404092).

TAF versus TDF at week 48

HBV DNA below 29 IU/mL	TAF	TDF
HBeAg-positive (Chan)	64%	67%
HBeAg-negative (Buti)	94%	93%

Median eGFR change in the HBeAg-negative trial: -1.8 (TAF) vs -4.8 mL/min (TDF)



Chan HLY et al. Lancet Gastroenterol Hepatol 2016;1:185-195 (PMID 28404091). Buti M et al. 2016;1:196-206 (PMID 28404092).

Choosing the drug: summary

Situation	Drug
Most patients	Entecavir, TDF or TAF
Kidney or bone disease, or at risk	TAF, or entecavir with the dose adjusted
Prior lamivudine resistance	TDF or TAF, not entecavir
Pregnancy	TDF (most data) or TAF
HIV	TDF or TAF with emtricitabine or lamivudine, inside the ART regimen
Decompensated cirrhosis	Entecavir or TDF. The US TAF label does not recommend Child-Pugh B or C
Wants a finite course, compensated	Pegylated interferon, in selected patients

Calculating PAGE-B

Variable	Category	Points
Age, years	16 to 29	0
	30 to 39	2
	40 to 49	4
	50 to 59	6
	60 to 69	8
	70 or older	10
Sex	Female	0
	Male	6
Platelets, x 10^9/L	200 or more	0
	100 to 199	6
	Under 100	9

52-year-old man

Started on TAF

Platelets 190

What is his PAGE-B score?

Points as printed in EASL 2025 Table 13. PAGE-B: Papatheodoridis GV et al. J Hepatol 2016.

Calculating PAGE-B

Variable	Category	Points
Age, years	16 to 29	0
	30 to 39	2
	40 to 49	4
	50 to 59	6
	60 to 69	8
	70 or older	10
Sex	Female	0
	Male	6
Platelets, x 10^9/L	200 or more	0
	100 to 199	6
	Under 100	9

52-year-old man

Started on TAF

Platelets 190

6 + 6 + 6 = 18

High risk: 18 or more

5-year HCC about 16 to 17%

Points as printed in EASL 2025 Table 13. PAGE-B: Papatheodoridis GV et al. J Hepatol 2016.

Calculate it yourself

PAGE-B	5-year HCC, derivation	5-year HCC, validation
9 or less	0%	0%
10 to 17	3%	4%
18 or more	17%	16%

- PAGE-B was built in treated patients; REACH-B was built in untreated patients
- The HBsAg-based model on the site is a 2026 research model, not validated

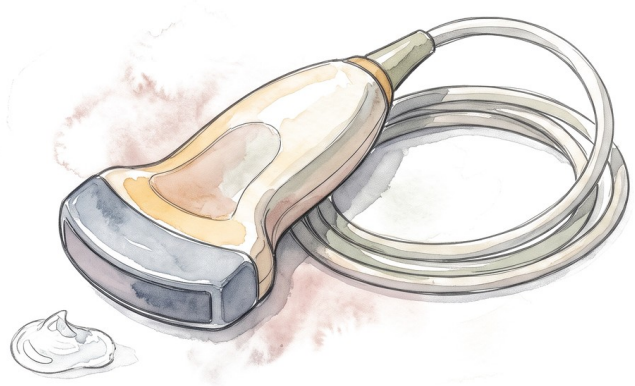
qaddodi.github.io/chb-hcc-risk

PAGE-B: Papatheodoridis GV et al. J Hepatol 2016. VA validation: Kim 2022.

Who gets surveillance?

Ultrasound and AFP every 6 months?

Patient	Surveillance
Compensated cirrhosis, Child-Pugh A	
Man from an endemic region, age 38, no cirrhosis	
Man from an endemic region, age 46, no cirrhosis	
Woman from an endemic region, age 46, no cirrhosis	
Family history of HCC	
PAGE-B score of 12	

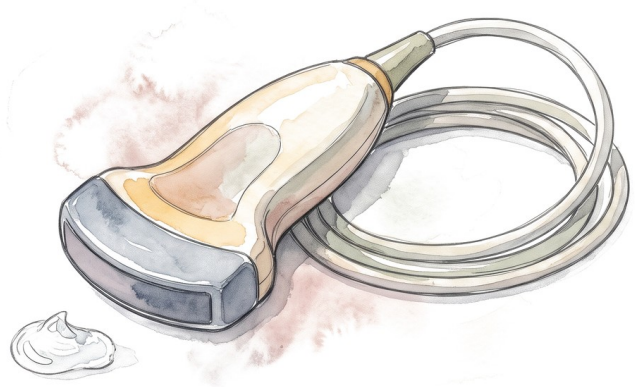


AASLD HCC practice guidance 2023, Table 1. Hepatology 2023;78:1922-1965, PMID 37199193.

Who gets surveillance?

Ultrasound and AFP every 6 months?

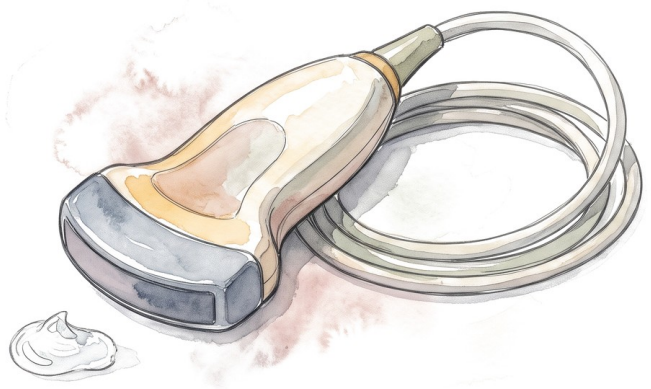
Patient	Surveillance
Compensated cirrhosis, Child-Pugh A	Yes
Man from an endemic region, age 38, no cirrhosis	No
Man from an endemic region, age 46, no cirrhosis	Yes
Woman from an endemic region, age 46, no cirrhosis	No
Family history of HCC	Yes
PAGE-B score of 12	Yes



AASLD HCC practice guidance 2023, Table 1. Hepatology 2023;78:1922-1965, PMID 37199193.

HCC surveillance: summary

- **Ultrasound and AFP every 6 months**
- Cirrhosis
- No cirrhosis, from an endemic region: men over 40, women over 50; people of African origin earlier
- Family history of HCC
- PAGE-B 10 or more
- Hepatitis D coinfection
- **Keep going after HBsAg loss if:**
- Cirrhosis or a family history of HCC, or HBsAg lost after 40 in men and after 50 in women



AASLD HCC guidance 2023, Table 1 (PMID 37199193); AASLD/IDSA 2025 R6 to R9 (PMID 41186418).

Risk of HBV reactivation

How risky is each drug if HBsAg is negative and anti-HBc is positive?

Drug	Reactivation risk
Rituximab or other anti-CD20	
Anthracyclines	
JAK inhibitors	
Prednisone 20 mg or more for 4 weeks or longer	
Anti-TNF	
Immune checkpoint inhibitors	
Methotrexate, azathioprine	

AGA guideline 2025, Table 6. Gastroenterology 2025;168:267-284, PMID 39863345. High above 10%, moderate 1 to 10%, low below 1%.

Risk of HBV reactivation

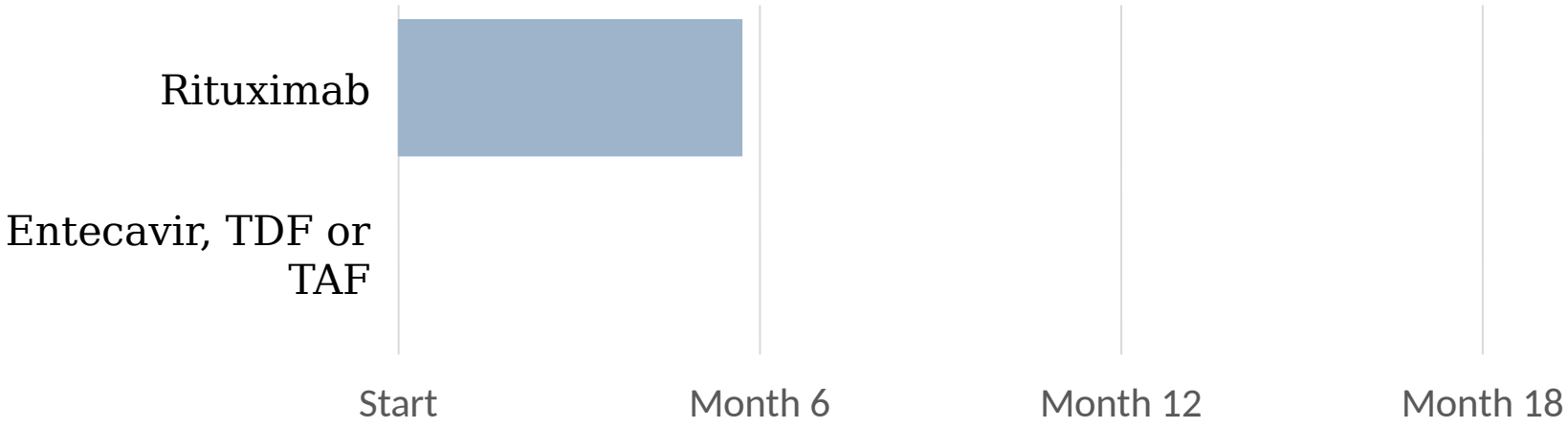
How risky is each drug if HBsAg is negative and anti-HBc is positive?

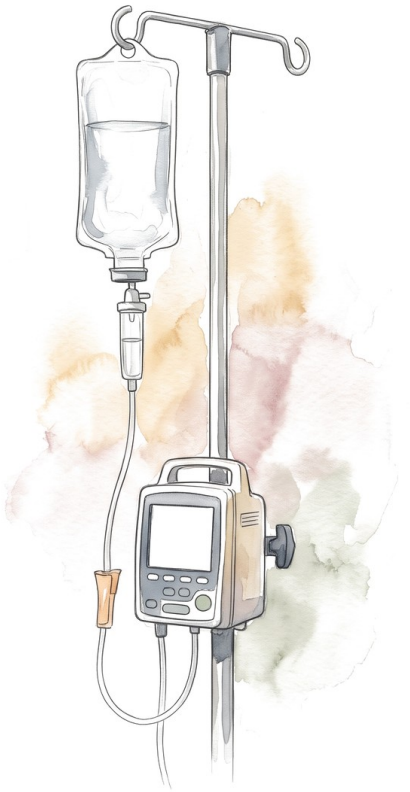
Drug	Reactivation risk
Rituximab or other anti-CD20	**High**
Anthracyclines	Moderate
JAK inhibitors	Moderate
Prednisone 20 mg or more for 4 weeks or longer	Moderate
Anti-TNF	Low
Immune checkpoint inhibitors	Low
Methotrexate, azathioprine	Low

AGA guideline 2025, Table 6. Gastroenterology 2025;168:267-284, PMID 39863345. High above 10%, moderate 1 to 10%, low below 1%.

Rituximab, six cycles

What do you give, and for how long?

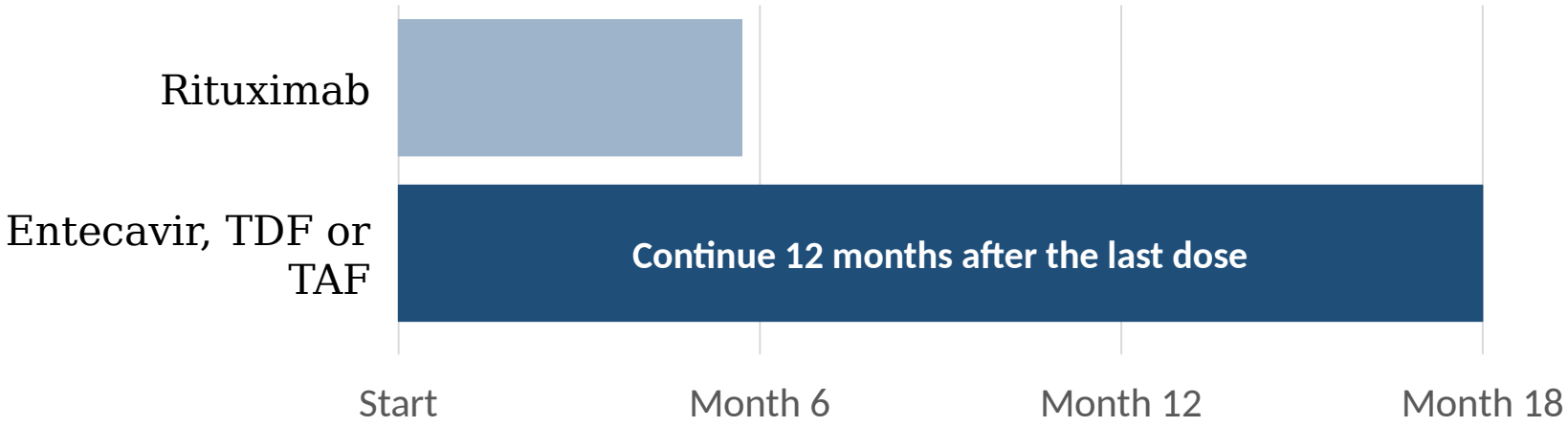


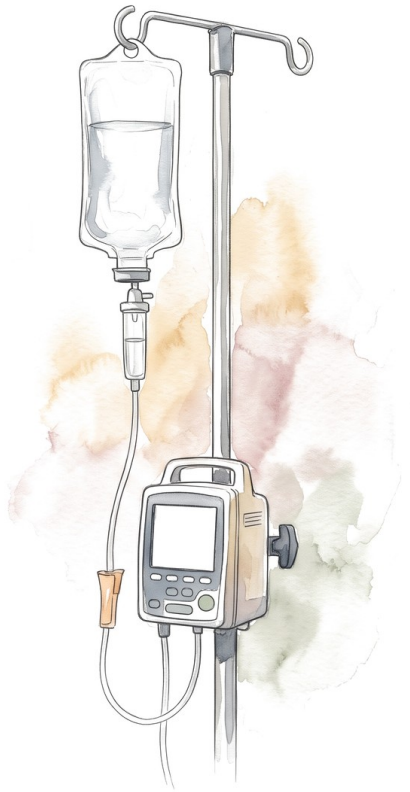


AGA 2025 (PMID 39863345); AASLD 2018 (PMID 29405329); EASL 2025 (PMID 40348683).

Rituximab, six cycles

What do you give, and for how long?

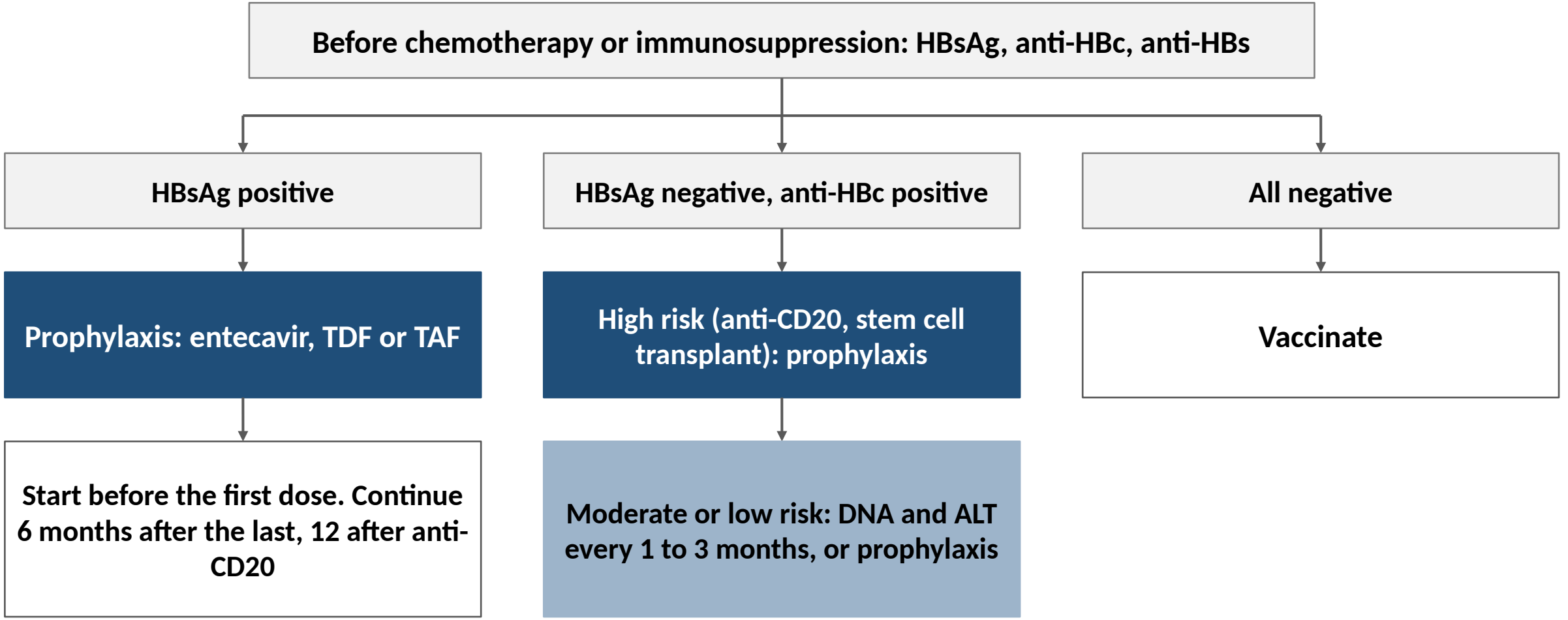




Start before the first dose. If you only monitor: DNA and ALT every 1 to 3 months

AGA 2025 (PMID 39863345); AASLD 2018 (PMID 29405329); EASL 2025 (PMID 40348683).

Before immunosuppression: summary



AGA 2025 (PMID 39863345); AASLD 2018 (PMID 29405329); EASL 2025 (PMID 40348683).

29-year-old woman, 20 weeks pregnant

HBsAg	HBeAg	HBV DNA	ALT
Positive	Positive	5,000,000 IU/mL	22 U/L

What do you do for her and for the baby?



29-year-old woman, 20 weeks pregnant

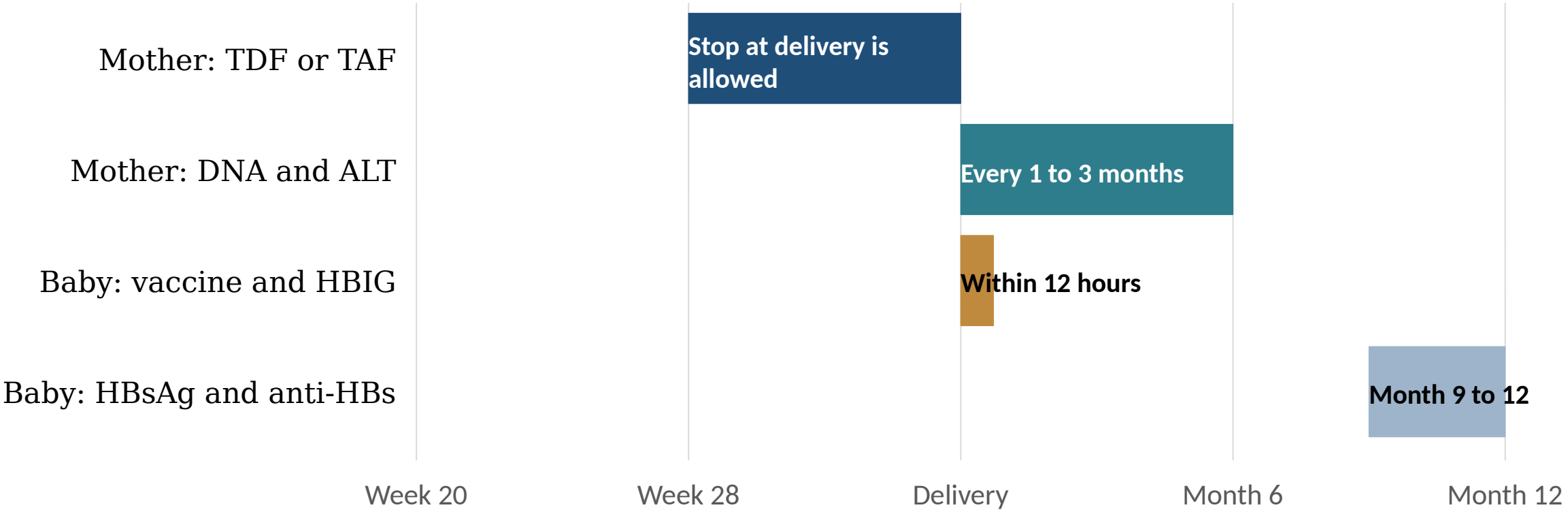
HBsAg	HBeAg	HBV DNA	ALT
Positive	Positive	5,000,000 IU/mL	22 U/L

What do you do for her and for the baby?

- TDF or TAF from week 28, because DNA is above 200,000 IU/mL
- Baby: hepatitis B vaccine and HBIG within 12 hours of birth



Her pregnancy and the baby's first year



AASLD/IDSA 2025 R1 (PMID 41186418); CDC perinatal hepatitis B overview, 2025.

Same woman, after delivery

She is on TDF. Can she breastfeed?

A. No, tenofovir in breast milk harms the baby

B. Yes

C. Only after the baby finishes the vaccine series

D. Only if her DNA is undetectable



AASLD 2018 (PMID 29405329); AASLD/IDSA 2025 (PMID 41186418).

Same woman, after delivery

She is on TDF. Can she breastfeed?

A. No, tenofovir in breast milk harms the baby

B. Yes

C. Only after the baby finishes the vaccine series

D. Only if her DNA is undetectable



AASLD 2018 (PMID 29405329); AASLD/IDSA 2025 (PMID 41186418).

Pregnancy: summary

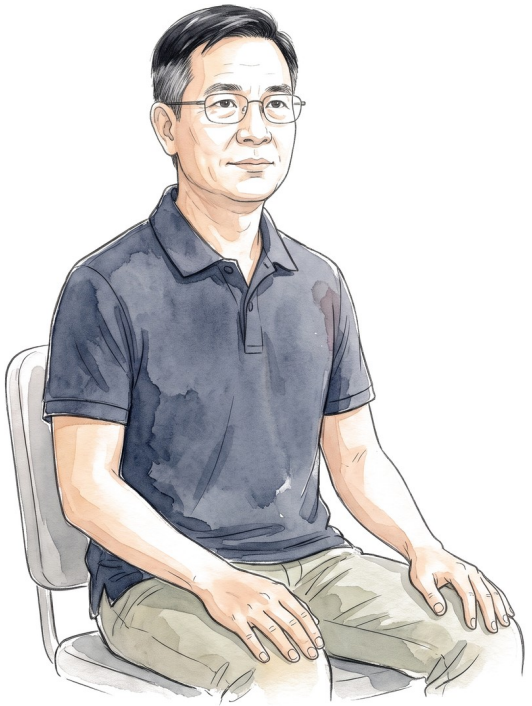
- Test HBsAg in every pregnancy
- HBsAg positive: check DNA and ALT
- **DNA above 200,000: TDF or TAF from week 28**
- Baby: hepatitis B vaccine and HBIG within 12 hours of birth, then finish the series
- Test the baby at 9 to 12 months: HBsAg and anti-HBs
- Breastfeeding is fine
- If she stops after delivery: DNA and ALT every 1 to 3 months for 6 months



AASLD/IDSA 2025 R1 (PMID 41186418); CDC 2023 (PMID 36893044); CDC perinatal hepatitis B guidance.

Same man, 57 years old, five years on TAF

Condition for stopping	Our patient	Met?
No cirrhosis or advanced disease	Stiffness 5.9 kPa	
HBV DNA undetectable for 2 years or more	Undetectable for 5 years	
HBsAg under 100 IU/mL	400 IU/mL	
No HIV or HDV	Both negative	
Reliable follow-up	Yes	

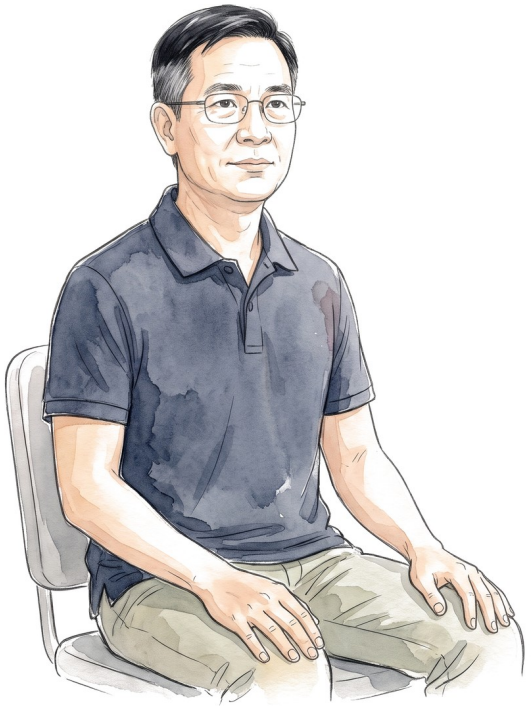


Can he stop?

AASLD/IDSA 2025 R5. Hepatology 2026;83:974-997, PMID 41186418.

Same man, 57 years old, five years on TAF

Condition for stopping	Our patient	Met?
No cirrhosis or advanced disease	Stiffness 5.9 kPa	Yes
HBV DNA undetectable for 2 years or more	Undetectable for 5 years	Yes
HBsAg under 100 IU/mL	400 IU/mL	**No**
No HIV or HDV	Both negative	Yes
Reliable follow-up	Yes	Yes



Can he stop?

No. Continue until HBsAg is lost

AASLD/IDSA 2025 R5. Hepatology 2026;83:974-997, PMID 41186418.

A different patient stops TAF

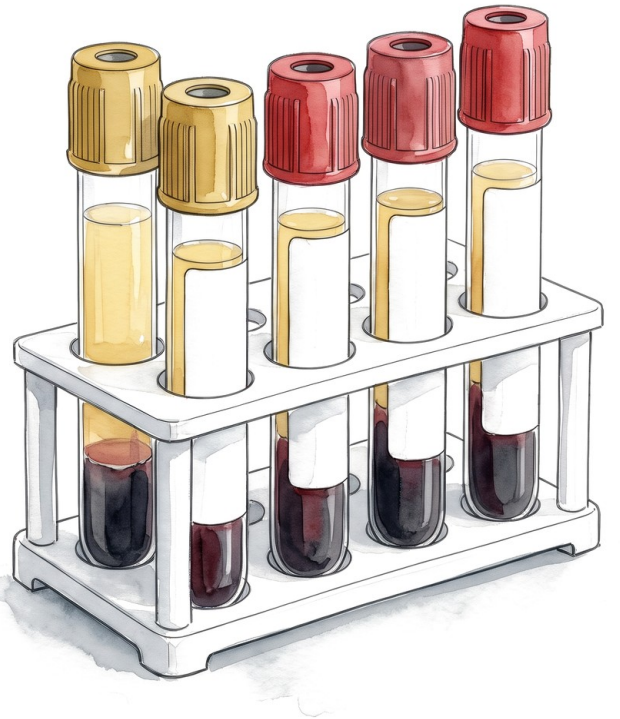
Which result means restart now?

A. ALT 1.5 times ULN

B. HBV DNA 10,000 IU/mL

C. HBsAg 200 IU/mL

D. Anti-HBs negative



AASLD/IDSA 2025 Table 3 (PMID 41186418).

A different patient stops TAF

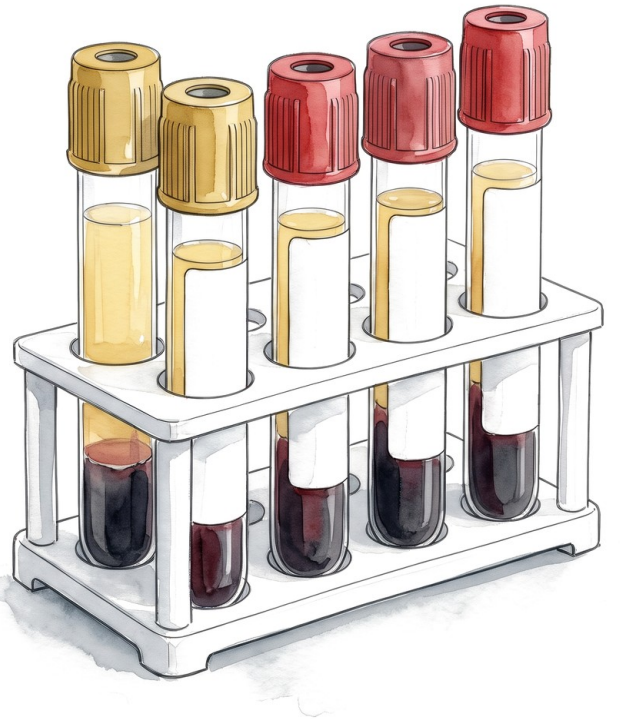
Which result means restart now?

A. ALT 1.5 times ULN

B. HBV DNA 10,000 IU/mL

C. HBsAg 200 IU/mL

D. Anti-HBs negative



AASLD/IDSA 2025 Table 3 (PMID 41186418).

Stopping: summary

- **Default: continue until HBsAg is lost**
- **Stop only if all of these:**
- No cirrhosis
- DNA undetectable for 2 years or more
- HBsAg under 100 IU/mL
- No HIV or HDV
- Reliable follow-up

After stopping

Months 0 to 6: DNA and ALT every 1 to 3 months

↓

Months 6 to 12: every 3 months

↓

After 12 months: every 3 to 6 months

Restart now: DNA 10,000 or more, ALT above 5 times ULN, bilirubin above 2.5 mg/dL, or decompensation

AASLD/IDSA 2025 R5 and Table 3 (PMID 41186418).

Approved in 2026

	Bepirovirsen (Hibsago)	Bulevirtide (Hepcludex)
For	Chronic hepatitis B	Chronic hepatitis D
FDA approval	20 October 2026	22 May 2026
Who	Adults without cirrhosis, on a nucleos(t)ide analogue 6 months or more, DNA suppressed, HBsAg 3,000 IU/mL or lower	Adults without cirrhosis or with compensated cirrhosis
Result	19% had HBsAg undetectable 24 weeks after stopping all treatment; 0% on placebo	Week 48: combined response 48% vs 2%
Warning	Liver reactions, low platelets	Severe exacerbation if stopped

FDA news releases 20 October 2026 (bepirovirsen) and 22 May 2026 (bulevirtide). B-Well 1 and 2, Hou, NEJM 2026.

Back to our patients

	Patient	Decision
1	Man, 34 to 57. HBeAg-positive, then HBeAg-negative with DNA 8,000 and ALT 48	Monitor at 34. Treat at 41 (age) or at 52 (grey zone). PAGE-B 18, surveillance. Do not stop with HBsAg 400
2	Woman, 61. Anti-HBc positive, HBsAg negative, rituximab	Entecavir, TDF or TAF before the first dose and for 12 months after the last
3	Woman, 29, 20 weeks pregnant. DNA 5,000,000	TDF or TAF from week 28. Vaccine and HBIG within 12 hours. Test the baby at 9 to 12 months



Questions?

X: @mfalmeqdadi

Slides (PDF)